Ibrahim Ibrahim

MSc Field Robotics Student | Erasmus Mundus Scholarship

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Education

MSc Erasmus Mundus in Intelligent Field Robotics | University of Girona / University of Zagreb

Year 1 - Girona, Spain | *September 2024– July 2025*

- Built a solid foundation in field robotics, focusing on motion planning, perception, and probabilistic localization (SLAM).
- Completed hands-on courses in manipulation, localization, and perception using ROS, Python, and C++.
- Gained experience in trajectory planning, visual SLAM, and autonomous navigation through lab-based projects.

Year 2 - Zagreb, Croatia | *September 2025 – June 2026*

- Specialized in multi-robot systems, aerial robotics, and human-robot interaction, emphasizing autonomy and coordination.
- Explored deep learning and AI techniques for perception and decision-making in robotics.
- Engaged in seminars and team projects involving control, sensing, and collaborative robotics.

BSc Computer Science | University of Pécs

Pécs, Hungary | *2021– 2024*

- Stipendium Hungaricum Scholarship recipient due to my exceptional academic performance in secondary school.
- Thesis topic: Machine Learning Force Fields for Transition Metal Complexes.
- Notable Modules: Data Analysis, Algorithms and Data Structures, Probability and Statistics, Control Technology.

Work Experience

Machine Learning Research Intern | CoE Marble

Zagreb, Croatia | *July 2025 - October 2025*

- Developed and evaluated visual anomaly detection models (Autoencoder U-Net, YOLO, EfficientAD, PatchCore) on offshore and wind-turbine inspection datasets.
- Implemented synthetic anomaly generation pipelines and created a custom underwater inspection dataset.
- Integrated explainability techniques (Grad-CAM, feature attribution maps) to interpret and visualize model decisions.
- Conducted benchmarking and literature review on state-of-the-art visual anomaly detection and explainable AI methods.

Automation & Data Engineer intern | DATelite

Pécs, Hungary | *October 2023 – April 2024*

- Established automation workflows using Airflow, streamlining data pipelines for enhanced
- Designed and executed ETL processes, developing Python functions for discrete ETL steps.
- Implemented statistical modelling and visualization plotting to derive meaningful insights from data.

Stipendium Hungaricum Mentor | SHMN

Pécs, Hungary | *July 2022 – July 2023*

- Welcomed international students to the university and provided up-to-date information about their studies and administrative tasks.
- Demonstrated good organization skills and the ability to plan ahead for 10-15 students.
- Demonstrated high adaptability and the ability to find alternative solutions.

Portfolio

Visit my online portfolio to explore selected projects in robotics, computer vision, and autonomous systems 190ibrahim.github.io/My-Portfolio Includes work on motion planning, SLAM, deep learning for perception, and embedded robotic systems.

Skills

- **Robotics & Automation:** Path planning, autonomous navigation, SLAM, Probabilistic robotics.
- **Artificial Intelligence & Machine Learning:** Deep learning, reinforcement learning, Explainable AI.
- **Embedded Systems & Programming:** C++, Python, ROS (Robot Operating System), MATLAB and SQL.
- **Mechatronics & Control Systems:** Motion planning, kinematics, and real-time control.
- **Tools:** Git & GitHub, Docker and Linux, Gazebo and Stonefish.

Languages

- Arabic: Native
- English: Fluent (B2)
- Hungarian: Beginner
- Spanish: Beginner